

Evolution of DMS Sensor Suite from SAE Level 1 to Level 4

DDAW: Driver Drowsiness and Attention Warning
AADW: Advanced Driver Distraction Warning

L2 already needs continuous driver monitoring

Key enabling technology and feature for high level of automation

DMS is the key for level 2 and level 3

	SAE Level 1	SAE Level 2	SAE Level 3	SAE Level 4	SAE Level 5
Regional regulation DDAW		Mandated in Europe from 2022			
Regional regulation DMS/AADW		Mandated for motor vehicles of categories M and N from 7 July 2024 for new types and from 7 July 2026 for all new vehicles Euro NCAP 5 Star 2025	Expected to be mandated. More complex systems required such as being able to monitor the biometric information (e.g., heart attack monitoring and alerting, etc.)		
Features	Lane-keep assist, no cruise control	Human hands-on at all times	Drivers are allowed to have hands-off for periods of time (AADW required to warn the drivers)	Fully autonomous driving in all road environments, restrictions for weather and extreme edge case events (e.g., tree fallen in road)	Fully autonomous driving
Technology Requirement	Torque sensor on steering wheels Lane keeping	Torque sensor on steering wheels Cameras (e.g., RGB cameras, ToF cameras, IR cameras, etc.) – not all needed, usually ToF and RGB are used. Radar, LiDAR	Torque sensor on steering wheels Cameras (e.g., RGB cameras, ToF cameras, IR cameras, etc.) – multiple needed Radar Wearable sensors such as ECG and EEG sensors – not expect to get widely adopted in this decade	Capacitive touch sensor Radar + LiDAR	

Regulatory landscape and SAE level. More details are in IDTechEx's research: *In-Cabin Sensing 2025-2035: Technologies, Opportunities, and Markets*

experience. Before exploring the technical details, it is worthwhile to understand what in-cabin sensing entails and why it matters in today's automotive landscape.

Car's hidden eyes and hand

Near-infrared (NIR) cameras are essential components in modern DMS. Paired with NIR LEDs, they project invisible light onto the driver's face. This light, undetectable to the human eye, reflects back to the camera, allowing the system to track head position, eye movements, and blinking. As NIR light is invisible, these systems function reliably in both day and night conditions, helping monitor driver alertness without distraction.

However, integrating NIR cameras into vehicles presents challenges. While higher-resolution cameras improve accuracy, the key hurdles lie in software and algorithms. With overlapping features across systems, companies focus on software that performs reliably across diverse

real-world conditions and avoids false readings, often depending on large datasets for training. Some systems now combine NIR with RGB cameras, which capture colour, while NIR provides detailed grayscale. This combination supports more features but adds complexity, as the software must process inputs from both sensor types.

Time-of-flight (ToF) cameras are another technology under exploration. Although less common in DMS, they are used in OMS for functions such as seat occupancy detection and gesture control. For instance, a second-row passenger could wave to lower a display screen. ToF's strength lies in capturing 3D depth data, but high costs—\$50-\$100 per unit—have hindered widespread adoption. Their

broader use will depend on either cost reductions or a clear return on the added value.

Radar sensors, particularly at 60GHz, offer a distinct advantage: they can detect through soft obstacles, such as blankets. This makes them ideal for identifying rear-facing infants in car seats, even when visibility is limited. Tesla, for example, uses in-cabin radar in its Model Y to monitor rear passengers, reinforcing its safety-first approach.

To monitor hand presence on the steering wheel, torque sensors have long been the standard. They detect whether the driver is applying force to the wheel, a feature integrated into power steering systems. However, they have limitations—light hand contact may go unnoticed, and they can be tricked by weights simulating hand pressure.

To improve reliability, some automakers now use capacitive touch sensors, which detect skin contact

Key Global Players

Seeing Machines (Australia): Software-focused
Mobileye: Global presence, software and systems
Cipia (Israel): Software provider
Sony: Sensor hardware
Magna: Tier 1 supplier involved in system integration